

ASTRAQ

DPM1 6x4 Diesel Prime Mover

Formal product specification



DPM1 6x4 Diesel Prime Mover

KEY DIMENSIONS

Application	6x4 prime mover / semi-trailer tractor
Length (mm)	7025
Width (mm)	2490
Height (mm) unloaded	3317
Wheelbase (mm)	3200 + 1400
GVM (kg)	25,000
GCM (kg)	90,000
Max speed (km/h)	100
Fuel tank (L)	700 + 240
Fifth-wheel height (mm)	1260 unloaded

FEATURES

- 6x4 diesel prime mover with one front steering axle and tandem rear drive axles.
- Right-hand drive cab configured for Australian prime mover applications.
- MC13.54-61 12.42L diesel engine: 400 kW / 1800 rpm and 2500 Nm / 950-1400 rpm.
- ZF 12TX2621TD 12-speed automated transmission with overdrive top gear.
- 25,000 kg GVM and 90,000 kg GCM.
- Tandem rear drive axles with inter-wheel and inter-axle differential locks.
- Dual-circuit compressed-air braking with EBS, ESC and hill-start assist.
- ACC, AEBS, LDW and reversing camera included for driver assistance.

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Vehicle parameters

WEIGHTS

Kerb weight (kg)	8,800
Kerb axle load - front (kg)	4,820
Kerb axle load - rear (kg)	3,980
GVM (kg)	25,000
GCM (kg)	90,000
Design axle load - front (kg)	7,000
Design axle load - rear (kg)	18,000

CAPABILITY

Max speed (km/h)	100
Minimum turning diameter (m)	14
Max gradeability (%)	≥34
Max parking grade (%)	≥20
Fuel tank capacity (L)	700 + 240

CABIN / COMFORT

Cab model	DPM1 RHD prime mover cab
Passenger capacity	2
Cab suspension	Four-point air suspension
Driver seat	Air suspension seat
Cab tilting	Electric + manual hydraulic
Multimedia	MP5
Air conditioning	Semi-automatic
Remote key	1 key

DIMENSIONS

Length (mm)	7025
Width (mm)	2490
Height (mm) unloaded	3317
Wheelbase (mm)	3200 + 1400
Wheel track - front (mm)	2041
Wheel track - rear (mm)	1842
Front overhang (mm)	1475
Rear overhang (mm)	950
Approach angle	19°
Departure angle	22°
Fifth-wheel height (mm)	1260 unloaded
Front / rear fitting radius (mm)	2280 / 1961

PRIME MOVER INTERFACE

Application interface	Rear fifth-wheel coupling
Semi-trailer connection	Included
Axle layout	1 front steering + 2 rear drive axles

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Powertrain and chassis

ENGINE

Model	MC13.54-61
Displacement (L)	12.42
Power rating	400 kW / 1800 rpm
Max torque	2500 Nm / 950-1400 rpm
Application	Heavy-duty diesel prime mover

TRANSMISSION AND CLUTCH

Transmission model	FAST
Forward ratios	16.688, 12.924, 9.926, 7.688, 5.895, 4.656, 3.655, 2.831, 2.174, 1.684, 1.291, 1.00
Reverse ratios	15.537, 12.033
Clutch model	430 push-type diaphragm spring clutch
Clutch operation	Electronically controlled pneumatic power assist

AXLES AND SUSPENSION

Front axle model	VPD71DS
Driving axle model	MCY13BES
Final drive ratio	3.08 / 3.36 / 3.70
Differential lock	Inter-wheel and inter-axle differential lock
Suspension model	Non-independent suspension
Front suspension	Parabolic springs, 3 leaves
Rear suspension	Air suspension, 8 air springs

TYRES AND FRAME

Rim	Front 22.5 x 9.0 aluminium alloy; rear 22.5 x 8.25 aluminium alloy
Tyres	Front 295/80R22.5; rear 11R22.5
Frame model	Trapezoidal frame with wide front and narrow rear
Main beam section (mm)	Slot type 270 x 85 x 8
Frame width (mm)	Front 942; rear 762

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Steering, braking, safety and electrics

STEERING

Model	Recirculating ball hydraulic power steering
Steering ratio	17-20:1
Max pump pressure (kPa)	20,000
Working pressure (kPa)	1,250
Steering way	Steering wheel

ELECTRICS / COMFORT

Operating voltage	24 V
Generator power	3 kW
Batteries	2 x 12 V / 240 Ah
Multimedia	MP5
Air conditioning	Semi-automatic
Remote key	1 key

BRAKING / SAFETY

Service brake	Dual-circuit compressed-air brake
Brake power assist type	EBS + ESC + HSA
Parking brake type	Spring stored energy air cut brake
Circuit type	Dual
Driver assistance	ACC + AEBS + LDW
Reverse camera	Included

PRODUCT POSITIONING

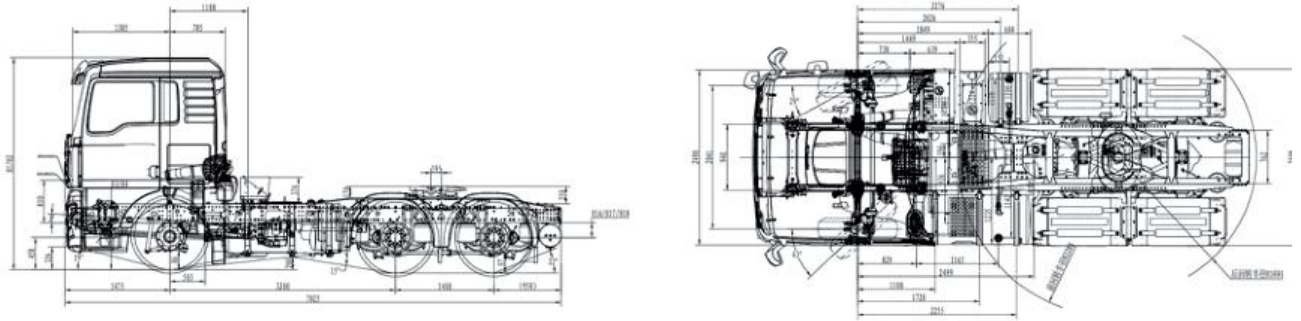
- DPM1 is configured as a 6x4 diesel prime mover for semi-trailer operation.
- The rear fifth-wheel coupling distinguishes DPM1 from a rigid cab chassis.
- The specification emphasises GCM, fifth-wheel height, tandem drive axles, air suspension, long-haul braking and driver-assistance functions.
- This document presents DPM1 as a product specification rather than a comparison or target configuration sheet.

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Chassis structure layout

STRUCTURE DRAWING



DPM1 6x4 prime mover chassis layout - side view and top view

LAYOUT HIGHLIGHTS

Axle configuration	1 front steering axle + 2 rear drive axles
Wheelbase (mm)	3200 + 1400
Application interface	Rear fifth-wheel coupling
Frame arrangement	Trapezoidal ladder frame
Rear suspension	Air suspension, 8 air springs
Fuel system	700 + 240 L

PACKAGING NOTES

- Side-view layout supports fifth-wheel, axle-spacing and rear overhang review.
- Top-view layout supports frame, driveline and component-packaging review.
- Chassis configuration is arranged for semi-trailer operation rather than direct rigid-body mounting.